

Exploring the Reaction Paths in the Consecutive Fe-Based FT Catalyst–Zeolite Process for Syngas Conversion

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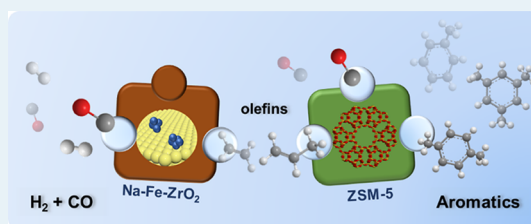
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ABSTRACT: Direct conversion of syngas into aromatics with tandem catalysts containing Fe-based Fischer–Tropsch synthesis (FTS) catalysts and zeolites has been attracting significant attention because of the great cost-efficiency. However, the interaction role between two active species and reaction paths are still unclear. We report here that a high yield of aromatics was achieved by serial reactions directly from syngas on a Na-Fe-ZrO₂/ZSM-5 tandem catalyst. Results point out significant impacts of zeolite in the catalytic system on performance including activity and selectivity. The role of the zeolite on reaction pathways about both CO conversion and aromatic formation has been systematically investigated by steady-state isotopic transient kinetic analysis and designed experiments. The presence of zeolite in the catalytic system promotes the CO hydrogenation rate by means of improving the iron carburization degree, possibly because of the enhanced CO adsorption and the formation of surface intermediates. It also enhances selectivity to aromatics by coupling CO to participate in the olefin transformation on ZSM-5. In addition, the formation rate of aromatics is proportional to the number of Lewis acidic sites, suggesting the dehydroaromatization route as the main mechanism of olefin aromatization. These findings provide some insights into the effect of ZSM-5 on reaction paths over the FTS catalyst–zeolite tandem catalyst system.

KEYWORDS: syngas, FT catalyst–zeolite process, mechanism, SSITKA, olefin aromatization



1. INTRODUCTION

Aromatics, one of the most important platform molecules in the current chemical industry, are widely used to produce a range of important petrochemicals including plastics, films, fibers, pesticides, nylon, and pigment dyes.^{1,2} Driven by the rapidly increasing consumption of benzene derivatives, the demand for aromatics is growing faster and faster.³ The aromatics come traditionally from petroleum refinery processes such as thermal cracking of crude oil and catalytic reforming of naphtha, which heavily rely on petroleum.^{4,5} In order to alleviate the dependence on fossil resources, some non-petroleum routes are developed to synthesize aromatics from alternative carbon sources.

Syngas that is derived from coal, natural gas, biomass, or organic waste could be converted into various hydrocarbons through Fischer–Tropsch (FT) synthesis.^{6,7} However, the distribution of products in conventional FT process is limited by Anderson–Schulz–Flory, leading to a quite low aromatic selectivity.^{8–10} Recently, researchers have proposed a composite catalyst that combines an oxide where CO is activated into C1 intermediates and a zeolite where the C–C coupling occurs to directly synthesize aromatics from syngas.^{11–16} For example, Cheng et al.¹⁷ reported a bi-functional catalyst composed of Zn-doped ZrO₂ nanoparticles and HZSM-5 zeolite, obtaining an 80% selectivity of aromatics. Nevertheless, the CO conversion is very low due to the thermodynamic equilibrium

limitation of CO hydrogenation into methanol, making it a great challenge for large-scale production.

Meantime, another route that involved the Fe-based Fischer–Tropsch synthesis (FTS) catalyst and zeolite is also put forward for the one-step conversion of syngas into aromatics, attracting wide attention by academia and industry.^{18–20} Zhao et al.²¹ designed a bifunctional catalyst, Na-Zn-Fe/ZSM-5, and applied in the CO hydrogenation reactions. Surprisingly, the aromatic selectivity achieved about 51% with CO conversion higher than 85% during the stable reaction stage. Low-carbon alkenes, which form over Fe-based catalyst, are regarded as intermediates and further transfer into zeolite for aromatization in this process. On the other hand, besides the direct alteration of final products, CO conversion is also influenced by the addition of zeolite.^{22–24} Cheng et al.²⁵ have reported that the chemical properties of the Fe-based FTS catalyst changed after adding the ZSM-5 catalyst because of the interaction between Fe-based catalyst and zeolite. However, the specific interaction and resulted influence are

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still uncertain. In addition, it is still in dispute about the mechanism of olefin aromatization over zeolites.^{26,27} Different from the generally believed hydrogen-transfer mechanism, Weber et al.²⁸ have demonstrated that olefins follow a dehydroaromatization pathway. Nevertheless, the main reactions and the nature of active sites over zeolite for the tandem process remain unclear. Consequently, systematic research is imperative to provide more insights into the reaction paths on the consecutive reactions.

Here, we combine Na-Fe-ZrO₂ with ZSM-5 as a tandem catalyst for the conversion of syngas to aromatics. By defining the relationship between product selectivity and CO conversion, the tandem reaction process is revealed. Besides, the interaction between two components, as well as reaction paths involved CO, are discussed by using steady-state isotopic transient kinetic analysis (SSITKA) analysis technology and deliberately designed experiments. Moreover, ZSM-5 is treated under NH₃ flow to change the acidic properties while maintaining the pore structure and Si/Al. By this method, the relationship between acidity and formation rates of different products is investigated to get a better understanding of the olefin aromatization paths.

2. EXPERIMENTAL SECTION

2.1. Materials. Ferric citrate and sodium citrate were purchased from Sigma-Aldrich Company Ltd. Zirconium dioxide was purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. ZSM-5 was purchased from Nankai University Catalyst Co. All chemicals were used as received without further purification.

2.2. Catalyst Preparation. **2.2.1. Synthesis of Na-Fe-ZrO₂ Catalyst.** Na-Fe-ZrO₂ was synthesized by a coimpregnation method. Briefly, 10.90 g of ferric citrate and 0.12 g of sodium citrate were dissolved into 50 mL of deionized water straight after adding 5 g ZrO₂ with stirring. After evaporating the excess water at 50 °C, the mixture was dried at 80 °C overnight and subsequently calcined in air at 400 °C for 5 h.

2.2.2. Treatment of ZSM-5 Catalyst. The parent ZSM-5 zeolite was treated in a NH₃ atmosphere. Namely, 3.0 g of ZSM-5 zeolite was calcined under NH₃ flow (50 mL min⁻¹) for 12 h at different temperatures ($T = 300, 400, 500, 700$ °C respectively). Then, the samples were purged in Ar flow (50 mL min⁻¹) for 1 h at a same temperature. The samples were denoted as ZSM-5- T .

2.2.3. Preparation of Na-Fe-ZrO₂/ZSM-5 Composite Catalysts. The tandem catalysts were prepared by physical mixing of Na-Fe-ZrO₂ and ZSM-5. Typically, the materials were separately pelleted into particles with 250–380 μ m, followed by granule mixing together with a mass ratio of Na-Fe-ZrO₂/ZSM-5 = 2 unless otherwise stated. Especially, the dual-bed mixing catalyst was placing Na-Fe-ZrO₂ above ZSM-5 with quartz between them.

2.3. Activity Measurements. The catalytic testing of CO hydrogenation was carried out in a continuous flow fixed-bed stainless steel reactor (i.d. 10 mm) equipped with a K-type thermocouple located at the center of the catalyst bed. Six hundred milligrams of catalyst pellets (200 mg Na-Fe-ZrO₂ mixed with 400 mg SiC or ZSM-5) were reduced in H₂ (20 mL min⁻¹) at 350 °C for 6 h under atmospheric pressure. Subsequently, the feed gas with a composition of CO/H₂/Ar (3/6/1, vol %) was introduced into the reactor after cooling to room temperature. Generally, the reaction was conducted under the condition of 2400 mL g⁻¹ h⁻¹, 3 MPa, and 280 °C

except for the labeled tests. Especially, in the experiments of altering space velocity, the same content of two active components, that is $m_{\text{Na-Fe-ZrO}_2}/m_{\text{Na-Fe-ZrO}_2/\text{ZSM-5}} = 1/1$, was adopted in order to better observe the change of product distribution. The space velocities were also altered for different catalysts to compare the product changes in similar CO conversions. The whole effluent gas after reactions was analyzed online using Agilent 7890B gas chromatographs. The selectivity of hydrocarbons was calculated excluding CO₂ as reported previously.²⁹ Additionally, the C₅⁺ refers to the remaining hydrocarbons after subtracting aromatics. The carbon balance was over 90% based on the result of detected hydrocarbon products.

The SSITKA experiments were performed on a fixed-bed quartz reactor. One hundred milligrams of sieved Na-Fe-ZrO₂ mixed with 200 mg of SiC or ZSM-5 (granule mixing in the range of 250–380 μ m) were weighed as catalysts. After in situ reduction in H₂/Ar flow (20/10 mL min⁻¹) at 350 °C for 6 h and subsequent cooling in Ar (30 mL min⁻¹), the feed was changed to the mixture of ¹²CO/H₂/Ar (0.75/7.5/16.75 mL min⁻¹, 0.18 MPa) for CO hydrogenation at 280 °C. The isotopic transients switch from ¹²CO/H₂/Ar to ¹³CO/H₂/Ar was conducted after 10 h on stream during a relatively stable reaction period. A separate mass spectrometer (MS) and a combined GC–MS were used to collect the signals of isotopic transient response and catalytic performance.^{30,31}

For C₂H₄ or C₂H₄/CO polymerization experiments, 400 mg of parent ZSM-5 was employed as a catalyst. C₂H₄/Ar (1/5, vol %) or C₂H₄/CO/Ar (1/1/4, vol %) were used as feed gases. Reaction conditions were as follows: 3 MPa, 280 °C, 2400 mL g⁻¹ h⁻¹. The products were analyzed in the same way as for CO hydrogenation reactions.

2.4. Catalyst Characterizations. Powder X-ray diffraction (XRD) patterns were recorded on a PW3040/60 X'Pert ProSuper (PANalytical) diffract meter equipped with Cu K α radiation source, operating at 40 kV, 40 mA. The scanning electron microscopy (SEM) images were obtained on a JSM-7800F microscope with an accelerating 3.0 kV voltage. In addition, SEM APREO microscope equipped with EDX analyzer was employed to analyze the elemental mapping of the catalysts. N₂ physical adsorption–desorption isotherms were also conducted on an ASAP 2460 Quantachrome instrument to acquire the surface area and pore size distribution of the samples. Before the measurement, the samples were degassed under vacuum for 5 h at 300 °C.

H₂ temperature-programmed reduction (H₂-TPR) measurements were conducted on a Micromeritics AutoChem II 2920 apparatus. Prior to the measurement, the samples were pretreated under Ar flow at 200 °C for 1 h. After cooling to 50 °C, 10 vol % H₂/Ar was introduced to pass through the sample while being heated to 500 °C with a ramping rate of 10 °C min⁻¹.

The room-temperature ⁵⁷Fe Mössbauer spectra were recorded on a Topologic 500A spectrometer. ⁵⁷Co(Rh) was used as a radioactive source moving in a constant acceleration. To determine the parameters including isomer shift (IS) and quadrupole splitting (QS), the obtained spectra were fitted with Lorentzian line shape using the MössWinn 4.0 pre-program. The IS values and Doppler velocities were calibrated by a standard α -iron foil.

Temperature-programmed desorption of NH₃ (NH₃-TPD) was performed on a Micromeritics AutoChem II 2920

chemisorber. Typically, the samples were pretreated in He flow at 500 °C for 30 min and then cooled down to 100 °C. Next, the gas was switched to NH₃ for adsorption. After saturated adsorption, the samples were treated in He flow and heated from 100 to 800 °C at a ramping rate of 10 °C min⁻¹. The signals of NH₃ desorption were recorded by a mass spectrometer system.

Infrared spectroscopy with pyridine adsorption (Py-IR) was carried out on a BRUKER Equinox 55 spectrometer in a transmission mode with a resolution of 4 cm⁻¹ at 4000–400 cm⁻¹ range. Before the test, the sample wafers were pretreated under dynamic vacuum at 500 °C for 30 min. Then, pyridine vapor was introduced to the sample for 5 min after cooling the sample to 50 °C. Last, the IR spectrum was collected after evacuation at 350 °C. The number of Brønsted and Lewis acid sites was estimated based on the integrated IR peak areas at 1540 cm⁻¹ band and 1450 cm⁻¹ band, respectively.

3. RESULTS AND DISCUSSION

3.1. CO Hydrogenation Performance over Na-Fe-ZrO₂/ZSM-5 Catalysts. The Na-promoted Fe-ZrO₂ catalyst is employed here as a Fe-based FTS catalyst. The weight ratio of Fe and Zr is 1/3.5, and the Na promoter content is 0.5 wt % (Figure S1). According to the XRD patterns and SEM images (Figures S2 and S3), the Fe species are mainly presented as Fe₂O₃ over fresh samples, which evenly distributed on ZrO₂ microspheres.³² The Fe species are further confirmed by Mössbauer spectroscopy, as shown in Figure S4. Among the patterns, two sextets with magnetic hyperfine fields of 50.7 and 48.8 T could be assigned to Fe₂O₃ of large particles, while the doublet with IS of 0.34 mm s⁻¹ and double pole splitting (QS) of 0.77 mm s⁻¹ could be ascribed to superparamagnetic Fe³⁺ in small particles of Fe₂O₃.^{33,34} On the other hand, ZSM-5 (Si/Al = 18) with a highly crystalline structure is used as the zeolite part of the tandem catalyst, as shown in Figure S5.

The CO hydrogenation reactions are tested over Na-Fe-ZrO₂, ZSM-5 and Na-Fe-ZrO₂/ZSM-5. For ZSM-5, the reaction shows no activity, and no products are detected. Table 1 illustrates the CO conversion rates and product

Table 1. CO Conversion Rates and Product Formation Rates Over Na-Fe-ZrO₂ and Na-Fe-ZrO₂/ZSM-5 for CO Hydrogenation Reaction^a

	Na-Fe-ZrO ₂	Na-Fe-ZrO ₂ /ZSM-5
$R_{\text{CO}}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.5	0.55
$R_{\text{CO}_2}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.19	0.2
$R_{\text{CH}_4}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.04	0.06
$R_{\text{C}_{2-4}}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.16	0.02
$R_{\text{C}_{2-4}^0}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.06	0.12
$R_{\text{C}_{5-6}^{\text{Aro}}}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.22	0.16
$R_{\text{Aro}}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.02	0.2
α	0.758	0.831

^aReaction conditions: 280 °C, 3 MPa, 2400 mL g⁻¹ h⁻¹.

formation rates over Na-Fe-ZrO₂ and Na-Fe-ZrO₂/ZSM-5. With regard to Na-Fe-ZrO₂, the CO conversion rate is 0.50 mmol g⁻¹ min⁻¹, whereas it increases to 0.55 mmol g⁻¹ min⁻¹ after mixing with ZSM-5, unambiguously showing that the presence of ZSM-5 enhances CO conversion. Apart from that, C₂₋₄ alkenes account for 32% in total hydrocarbons over Na-Fe-ZrO₂, but the selectivity decreases to only 4% after

combining with ZSM-5. In contrast, the aromatic selectivity grows from 5 to 36% (Figure 1a and Table S1), and the

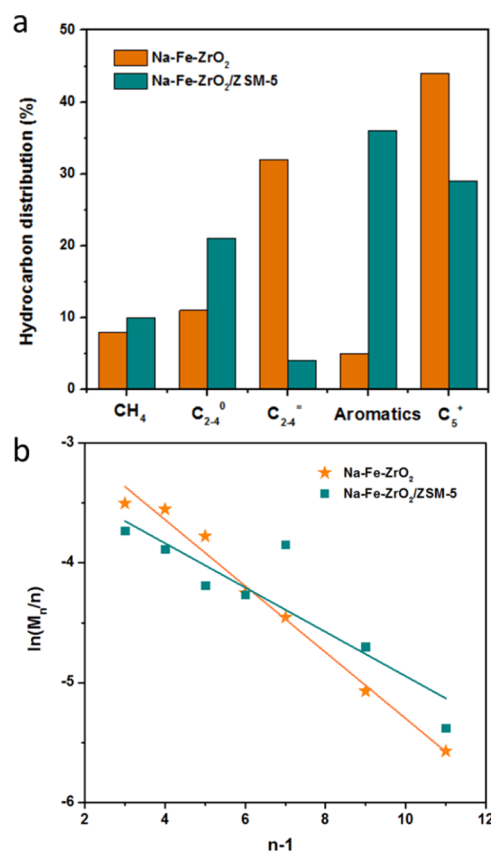


Figure 1. (a) Hydrocarbon distribution and (b) carbon chain growth probability over Na-Fe-ZrO₂ and Na-Fe-ZrO₂/ZSM-5 catalysts for CO hydrogenation reaction. Reaction conditions: 280 °C, 3 MPa, 2400 mL g⁻¹ h⁻¹.

formation rate of aromatics significantly rises even from 0.02 to 0.20 mmol g⁻¹ min⁻¹. The estimated carbon chain growth probability increased from 0.758 on Na-Fe-ZrO₂ to 0.831 on Na-Fe-ZrO₂/ZSM-5 as demonstrated in Figure 1b, indicating the products move to longer chain hydrocarbons with the tandem reaction.

The selectivity of various products is measured at different CO conversion induced by varying synthesis gas space velocity over these catalysts, to elucidate the reaction pathways of the tandem reactions (Figure 2). On the Na-Fe-ZrO₂, it is found that the product distribution is rather similar up to the CO conversion of 70%. It suggests that all of the products including light olefins, paraffins, and trace aromatics are the primary products, although it follows the Anderson–Schulz–Flory product distribution. It indicates that the surface reaction steps are relatively fast, and all of the intermediates form a product pool. However, the light alkane selectivity slightly increases while the aromatic selectivity decreases at the very high CO conversion, most likely as a result of enhanced WGS activity and increasing H₂/CO ratio (Table S2).³⁵

On a Na-Fe-ZrO₂/ZSM-5 catalyst, the trend in product selectivity is different from that of Na-Fe-ZrO₂. First, the selectivity to C₂–C₄ olefins decreases, whereas the selectivity to aromatics increases with increasing CO conversion. It suggests that the tandem reaction of C₂₋₄ olefins takes place, leading to the formation of aromatics and hydrogen through

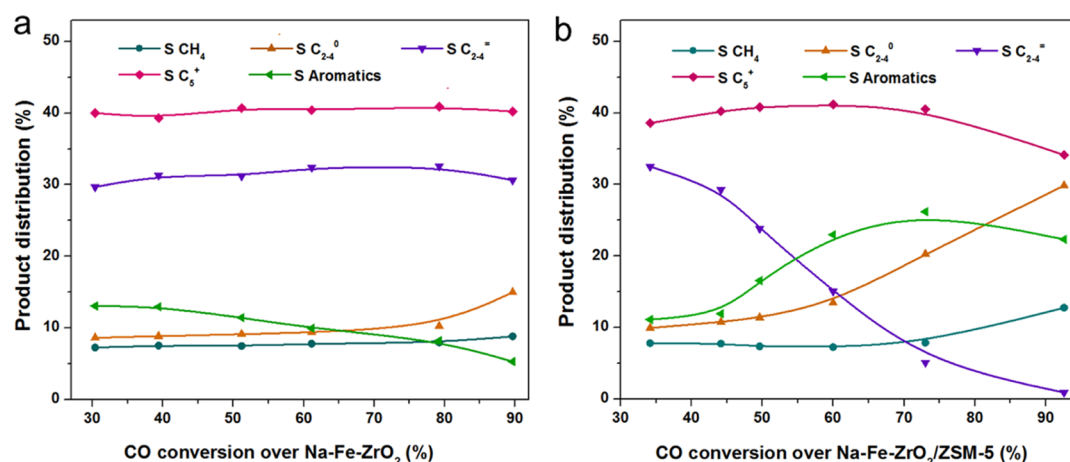


Figure 2. Product distribution as a function of CO conversion on (a) Na-Fe-ZrO₂ and (b) Na-Fe-ZrO₂/ZSM-5 for CO hydrogenation reaction. Reaction conditions: 280 °C, 3 MPa, 1000–9400 mL g^{−1} h^{−1}, $m_{\text{Na-Fe-ZrO}_2}/m_{\text{Na-Fe-ZrO}_2/\text{ZSM-5}} = 1/1$.

Table 2. SSITKA Parameters Over ZSM-5, Na-Fe-ZrO₂, and Na-Fe-ZrO₂/ZSM-5 Catalysts^a

	$X_{\text{CO}}/\%$	$R_{\text{CO}}/\mu\text{mol g}^{-1} \text{s}^{-1}$	$R_{\text{CH}_4}/\mu\text{mol g}^{-1} \text{s}^{-1}$	$\tau_{\text{CO}}/\text{s}$	$N_{\text{CO}}/\text{mmol g}^{-1}$	$\tau_{\text{CH}_4}/\text{s}$	$N_{\text{CH}_4}/\text{mmol g}^{-1}$
ZSM-5	0	0	0	12.0	34	0	0
Na-Fe-ZrO ₂	7.6	0.42	0.06	13.3	69	35.4	2.2
Na-Fe-ZrO ₂ /ZSM-5	9.4	0.52	0.08	30.2	152	35.7	3.0

^aReaction conditions: H₂/CO = 10, 280 °C, 0.18 MPa, 15,000 mL g^{−1} h^{−1}.

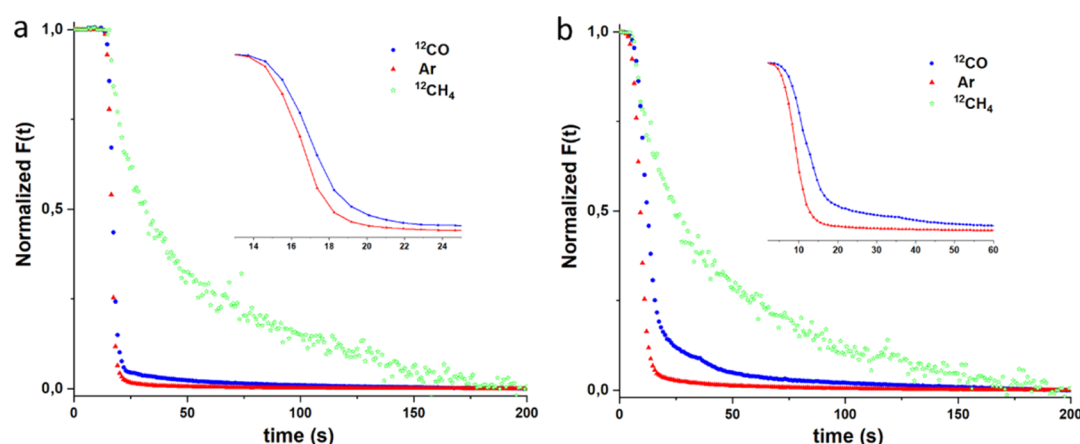


Figure 3. Normalized fitting curve of SSITKA experiments over (a) Na-Fe-ZrO₂ and (b) Na-Fe-ZrO₂/ZSM-5 catalysts. Reaction conditions: H₂/CO = 10, 280 °C, 0.18 MPa, 15,000 mL g^{−1} h^{−1}.

aromatization. Because the long contact time is more beneficial for the slow reaction, this change also implies that the formation of light olefins from CO hydrogenation is a quicker process than their conversion to aromatics. Subsequently, with the aromatization reaction becoming significant, the selectivity of light alkanes climbs gradually because of the hydrogen-transfer reaction or the hydrogenation with the H₂ from dehydroaromatization. Notably, at the high CO conversion, the H₂/CO ratio is possibly higher due to the promoted WGS reaction and the hydrogen generation from aromatization reactions, resulting in a reduced selectivity of aromatics and higher selectivity of alkanes in this tandem reaction.

3.2. Identification of CO Reaction Paths. SSITKA allows measuring the surface kinetic parameters, including the intrinsic surface residence time of CO and intermediates leading to CH_x (τ), as well as the surface concentrations of CO and CH_x (N) in the reaction process.^{36–38} Here, the SSITKA

is employed to investigate the impact of ZSM-5 addition on the surface reaction kinetic parameters. In order to exclude the CO adsorption over porous ZSM-5, the switch is conducted over pure ZSM-5 first. It shows that there is no activity at reaction condition, but there is CO adsorption, about 34 mmol g^{−1} (Figure S6). Apart from that, the addition of ZSM-5 to the Na-Fe-ZrO₂ catalyst enhances the catalytic activity with the CO conversion rate increasing from 0.42 to 0.52 $\mu\text{mol g}^{-1} \text{s}^{-1}$, which is in consistency with reaction results under high pressure (Table 2). Because of the high H₂/CO of feed gas employed in SSITKA experiments, significant alkane amount accompanying low olefin selectivity are observed. By quantitative calculation of the normalized transient response of Ar, CO, and CH₄ over separated Na-Fe-ZrO₂ and Na-Fe-ZrO₂/ZSM-5 in Figure 3, it is found that both τ_{CO} and N_{CO} are much higher over Na-Fe-ZrO₂/ZSM-5. After excluding the adsorption of the ZSM-5 part (Na-Fe-ZrO₂/ZSM-5 = 2, 2 ×

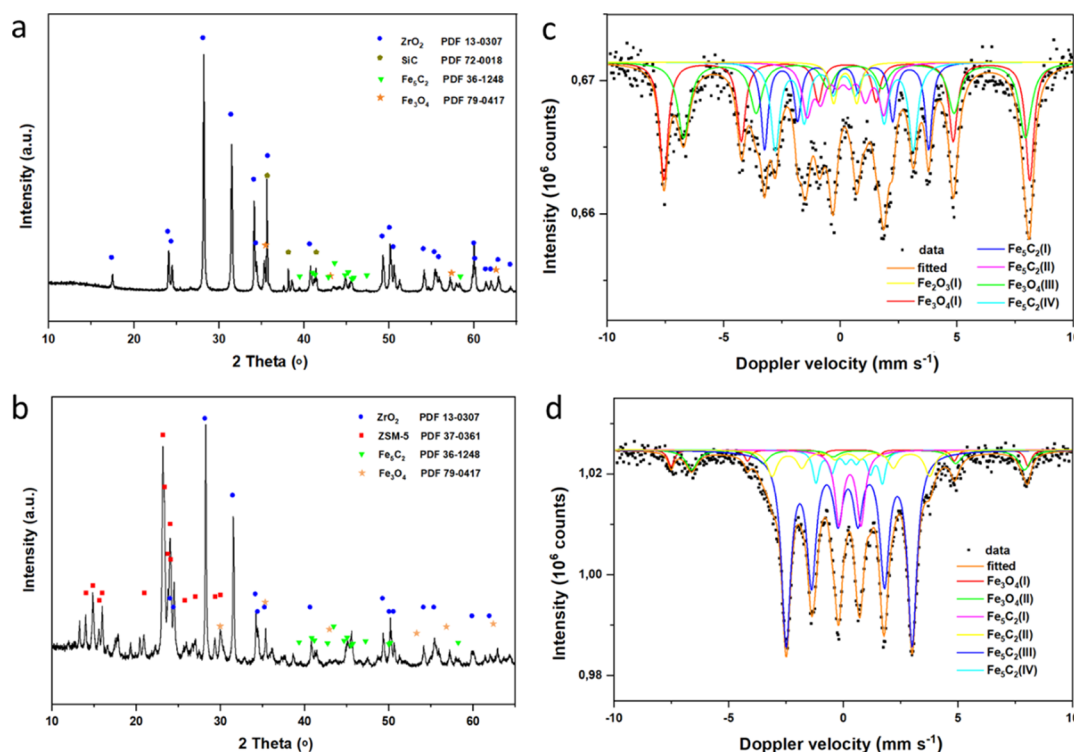


Figure 4. XRD patterns of (a) Na-Fe-ZrO₂ and (b) Na-Fe-ZrO₂/ZSM-5 after CO hydrogenation reaction; Mössbauer spectra of (c) Na-Fe-ZrO₂ and (d) Na-Fe-ZrO₂/ZSM-5 after CO hydrogenation reaction.

34 = 68 mmol g⁻¹), the surface CO concentration is still 15 mmol g⁻¹ higher than that on Na-Fe-ZrO₂. Additionally, the CH_x intermediate species over two catalysts are analyzed. For Na-Fe-ZrO₂/ZSM-5, the τ_{CH_x} and N_{CH_x} are 35.7 s and 3.0 mmol g⁻¹, which is greater than that on the Na-Fe-ZrO₂, about 35.4 s and 2.2 mmol g⁻¹, respectively. The results suggest a clear synergistic effect for Na-Fe-ZrO₂/ZSM-5 compared to the separate Na-Fe-ZrO₂ and ZSM-5, which causes a higher ability to form intermediate species leading to the products. Moreover, Figure 3 indicates that the shape of ¹²CO/¹³CO response curve is different. In contrast to the ¹²CO/¹³CO response curve on Na-Fe-ZrO₂, the one on Na-Fe-ZrO₂/ZSM-5 can be divided into two parts. In addition, the logarithm curves of two catalysts also display different shapes (Figure S7), indicating that CO might be involved in another different reaction pool over the tandem catalyst. Undoubtedly, CO is mainly activated and hydrogenated over the Fe-based catalyst, but the other reaction pool caused by the addition of ZSM-5 is still unexplored. In order to investigate the essential reason of improved activity, both the change of CO hydrogenation and the potential reaction over ZSM-5 are discussed subsequently.

3.2.1. CO Hydrogenation over Tandem Catalyst. Because the chemical state of Fe is directly related to CO hydrogenation, the differences of Fe species over both Fe-based catalyst and the tandem catalyst (250–380 μm particle mixing) are analyzed to determine whether ZSM-5 has any effects. First, the H₂-TPR over separated Na-Fe-ZrO₂ and Na-Fe-ZrO₂/ZSM-5 are conducted to check the reduction ability of Fe species. Both have three peaks appearing in Figure S8, which attribute the reduction of Fe₂O₃ to Fe₃O₄ and further reduction to FeO and metallic iron, respectively.

The states of Fe after 10 h reaction over the two catalysts are also evaluated. Based on the XRD patterns in Figure 4a,b, a peak at 44.2° that belongs to Hägg iron carbides (Fe₃C₂) is

observed except for the apparent peaks of Fe₃O₄ phase, indicating that iron species exist in the form of iron carbide and magnetite on both catalysts.³⁹ That means the iron species are in situ carburized in the process of CO hydrogenation. The EDS mapping results (Figure S9) indicate that the carbon element exists uniformly with the iron element. Further refining the XRD spectrum and analyzing the content of each component, it exhibits that the percent of iron carbides among total iron species is higher over Na-Fe-ZrO₂/ZSM-5 than sole Na-Fe-ZrO₂.

Given that Mössbauer spectroscopy has a better sensitive quantitative analysis for Fe species, it is employed here to detect the precise state of Fe species and their corresponding contents. In consistent with the XRD results, the magnetite and iron carbides species are found (Figure 4c,d). Besides, the fraction of iron carbide is much larger over the tandem catalyst (88.4%) comparing with that over separated Na-Fe-ZrO₂ (51.5%) (Table S4). This enhanced carburization is inferred to be caused by the massive adsorbed CO on ZSM-5 and a more intense mass transfer of carbon-containing intermediates between Na-Fe-ZrO₂ and ZSM-5.⁴⁰ In order to verify this hypothesis, the iron states over Na-Fe-ZrO₂ in dual-bed mode after reaction are also detected (Figure S10). As expected, the iron carbides occupy 87% in total iron species, much larger than that of separated Na-Fe-ZrO₂, but just a little bit smaller than Na-Fe-ZrO₂/ZSM-5 due to the slightly weak diffusion rate.

The iron carbides species are believed to be beneficial for CO hydrogenation activity and selectivity toward olefins.^{41,42} That is to say, the enhanced iron carburization is beneficial for the activity improvement and formation of olefins over the Fe-based catalyst.

3.2.2. CO and Olefins Coupling Mechanism. SSITKA results reveal that ZSM-5 introduces an additional CO reaction

Table 3. Catalytic Performance of ZSM-5 for C₂H₄ Coupling Reactions at Different Gas Atmospheres^a

	X _{C₂H₄} /%	X _{CO} /%	S _{C₃₋₅} ^c /%	S _{C₆₋₅} ^c /%	S _{C₆₋₅-Aro} ^c /%	S _{Aro} ^c /%
C ₂ H ₄ /Ar	81	— ^b	31	28	20	21
C ₂ H ₄ /CO/Ar	84	7	14	23	28	35

^aReaction condition: 3 MPa, 280 °C, 2400 mL g⁻¹ h⁻¹. ^bNo corresponding value

pool. Because olefins are acting as intermediates, CO may couple with olefins over ZSM-5. To test this hypothesis, coupling reactions are carried out over the ZSM-5 catalyst with C₂H₄/Ar or C₂H₄/CO/Ar as feed gases. As shown in Table 3, the C₂H₄ conversion is 81% in a sole C₂H₄ atmosphere, whereas it becomes 84% in the mixture gas of C₂H₄ and CO, and the CO conversion is about 7%. With respect to the product distribution, it is found that the C₃₋₅ selectivity is 31% and aromatic selectivity is 21% over a pure olefin atmosphere. However, the selectivity of low-carbon olefins decreases to 14%, whereas the selectivity of aromatics increases to 35% after adding CO in the feed gas. This indicates that CO could involve the coupling reactions of C₂H₄ and other low-carbon olefins to form aromatics over ZSM-5 in the process, which is consistent with the SSITKA results. Besides, the aromatic distribution in the two reactions over both feed gas atmospheres reveals that the proportion of aromatics with different carbon numbers in the mixture gas feed shows a similar trend with that of syngas conversion over the tandem catalyst, as shown in Figure 5. This phenomenon further implies the incorporation of CO into aromatics by coupling olefins over ZSM-5.

Because the SSITKA results demonstrate that there is a relative strong interaction of CO on HZSM-5 with a surface concentration of 34 mmol g⁻¹ at the reaction conditions, FTIR measurement is employed to understand the CO adsorption properties over ZSM-5. As Figure S11 shows, there are three peaks existing after He purge. The peak centered at 2135 cm⁻¹ corresponds to CO physical adsorption (condensation) in the pores of ZSM-5.⁴³ As for the other two peaks, Aréan et al.⁴⁴ have assigned them as the C–O stretching mode of monocarbonyl where adsorbed CO molecule interacts through the carbon atom (2175 cm⁻¹) and the oxygen atom (2111 cm⁻¹) with the cation in ZSM-5, respectively. Szanyi and Paffett⁴³ have also found that CO would adsorb onto the protons of the Si(OH)Al groups and extra-framework aluminum species. Furthermore, Brand et al.⁴⁵ studied by computational calculations that the interaction with proton could decrease the CO bond length. This part of adsorbed carbonyl compounds might react with unsaturated bonds, which easily transforms into oxygenates and converts into aromatics.⁴⁶ For instance, CO would form carbonyl compounds and react with olefins to form methyl-2-cyclopentene-1-ones intermediates that then transfer into aromatics according to the study of Chen et al.⁴⁷ Therefore, based on the above results, it is deduced that the presence of zeolite would facilitate the coupling reaction of CO with olefins into aromatics in this tandem process.

Considering all of the results here, it can be concluded that the presence of zeolite in the tandem catalyst has threefold effects. In addition to the introduction of serial reaction that transforms olefin intermediates to aromatics, it also enhances the carbonization of Fe catalysts. Moreover, CO molecules would also form carbonyl groups over zeolite and then incorporate into aromatics by coupling with olefins, participating in hydrocarbon pools in this tandem process, which not

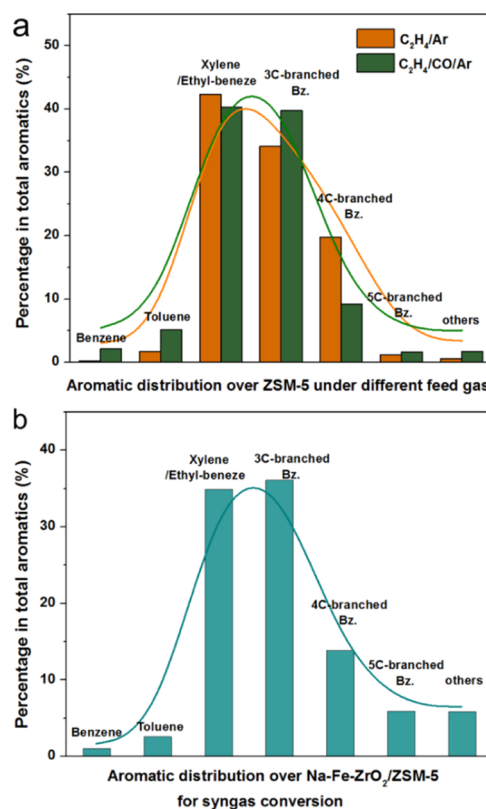


Figure 5. (a) Aromatic distribution over ZSM-5 for C₂H₄ coupling reactions at different reaction atmospheres. (b) Aromatic distribution over Na-Fe-ZrO₂/ZSM-5 for CO hydrogenation. Both experiments were conducted under the conditions of 280 °C, 3 MPa and 2400 mL g⁻¹ h⁻¹.

only promotes the CO conversion rate but also enhances the aromatic formation.

3.3. Reaction Paths of Olefin Aromatization. Besides the above reaction paths, the olefin aromatization route over the acidic sites of ZSM-5 is still in debate. In order to elucidate the main reaction pathways, different ZSM-5 zeolites are used to establish the relationship between acidic properties and product distribution. Here, NH₃ is used to treat ZSM-5 at different temperatures to alter the acidic properties while keeping the pore structure unchanged.⁴⁸

The XRD patterns shown in Figure S12a reveal that the crystallographic structures are quite similar among this series of treated ZSM-5 samples, which all exhibit the characteristic of MFI structure.⁴⁹ In addition, the analogous N₂ adsorption/desorption isotherms and surface area in Figure S12b and Table S5 clearly show that the typical microporous structure of ZSM-5 remains in original shape. Only a slight change in the mesoporous pore area is discovered. The specific morphologies of these treated zeolites are also tracked by SEM images, as displayed in Figure S13. All of them own identical pristine crystals with aggregated small particles. These results indicate that the ZSM-5 structures are almost the same after treatment.

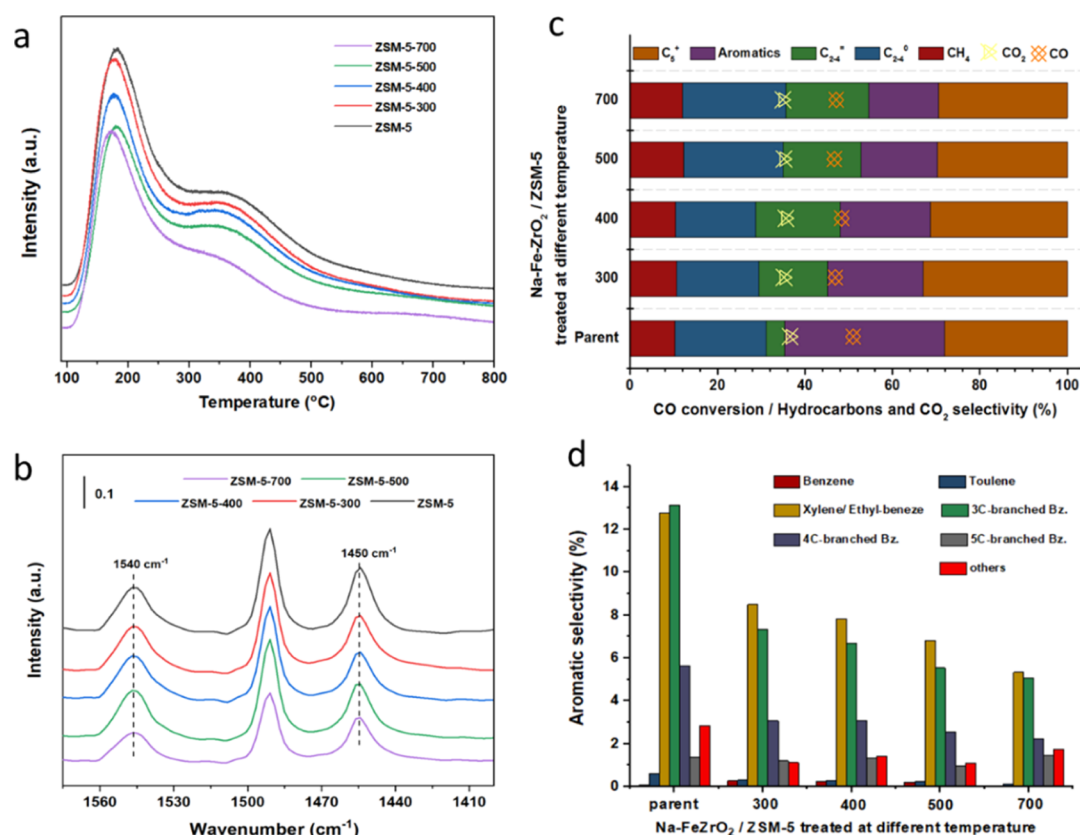


Figure 6. (a) NH₃-TPD curves and (b) Py-adsorption FTIR curves of ZSM-5 zeolites with treating NH₃ at different temperatures. (c) Product distribution and (d) aromatic selectivity over the Na-Fe-ZrO₂/ZSM-5-*T* catalysts. Reaction conditions: 280 °C, 3 MPa, 2400 mL g⁻¹ h⁻¹.

Table 4. CO Conversion Rates and Product Formation Rates Over Na-Fe-ZrO₂/ZSM-5-*T* Catalysts^a

	Na-Fe-ZrO ₂ /ZSM-5	Na-Fe-ZrO ₂ /ZSM-5-300	Na-Fe-ZrO ₂ /ZSM-5-400	Na-Fe-ZrO ₂ /ZSM-5-500	Na-Fe-ZrO ₂ /ZSM-5-700
$R_{\text{CO}}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.55	0.506	0.522	0.504	0.508
$R_{\text{CO}_2}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.203	0.18	0.187	0.179	0.179
$R_{\text{CH}_4}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.056	0.054	0.054	0.062	0.061
$R_{\text{C}_{2-4}^+}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.023	0.08	0.101	0.089	0.096
$R_{\text{C}_{2-4}^0}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.115	0.095	0.095	0.115	0.12
$R_{\text{C}_{5-7}^+}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.155	0.167	0.163	0.15	0.15
$R_{\text{Aro}}/\text{mmol g}^{-1} \text{ min}^{-1}$	0.2	0.11	0.107	0.088	0.081
H/C ratio	2.28	2.37	2.36	2.44	2.44

^aReaction conditions: 280 °C, 3 MPa, 2400 mL g⁻¹ h⁻¹.

In contrast to the physical properties, the acidic properties change significantly upon NH₃ treatment. The total acid amount obviously declines with increasing the treatment temperature based on the NH₃-TPD curves. By peak fitting and area integration, it reveals that both the strong acid sites and the weak acid sites reduced (Figure 6a,b).⁵⁰ Py-IR spectra are then collected to check the change of the number of Brønsted and Lewis acid sites. Under low treatment temperature (300 and 400 °C), the density of Lewis acid sites decreases from original 0.306 mmol g⁻¹ to 0.238 and 0.204 mmol g⁻¹, whereas the density of Brønsted acid sites just varies little from original 0.364 mmol g⁻¹ to 0.357 and 0.333 mmol g⁻¹. However, a large reduction of Brønsted acid sites but tiny changes of Lewis acid sites emerge after NH₃ treatment at 700 °C, getting values about 0.247 and 0.199 mmol g⁻¹, respectively.

3.3.1. Catalytic Performance over Na-Fe-ZrO₂/ZSM-5-*T* Catalysts. As there are distinctive reactions taking place at different acidic sites, the acidity properties would be a crucial factor for product distribution. The catalytic performance over the tandem catalysts that contain treated ZSM-5 are further performed and compared, as shown in Figure 6c,d and Table 4. The CO conversion and CO₂ selectivity are similar among these catalysts. However, the product distribution is quite different. For parent ZSM-5, the selectivity of C₂₋₄ alkenes is low (4%) and that of aromatics is high (36%) because of the effective aromatization. However, the aromatic selectivity drops to only 15% with treated ZSM-5-700. Especially, the selectivity toward olefins increases significantly with that of alkanes minor declining over the catalysts composed of ZSM-5-300 or ZSM-5-400. The selectivity of low-carbon paraffin increases over the catalyst composed of severer treated ZSM-5-700. The varied tendency seems to be similar to the changes in

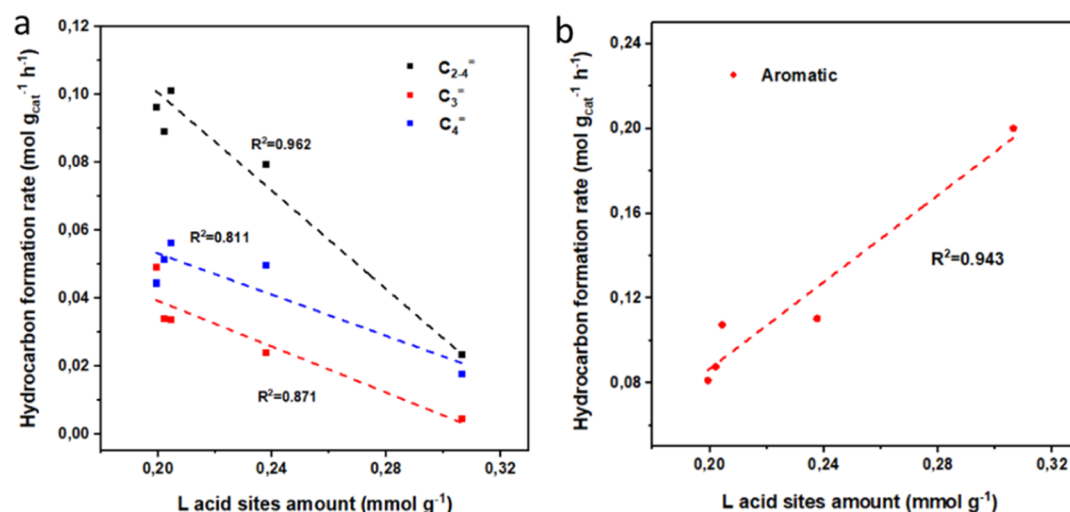


Figure 7. Relationship between Lewis acid amounts and (a) olefin or (b) aromatic formation rates.

acidic sites that is quantified by Py-IR, implying the potential reaction mechanism involved in this process.

3.3.2. Mechanism of Olefin Aromatization. It is generally believed that the hydrogen transfer reaction over Brønsted acid sites is responsible for the formation of aromatics from alkenes.^{1,51} With this mechanism, the change in light alkanes selectivity would be in proportional to that of aromatics,⁵² keeping H/C of total products as a certain value. However, with a little decrease of Brønsted acid sites from parent ZSM-5 to ZSM-5-300, the selectivity of aromatics changed from 36 to 22% accompanied by a tiny decrease of alkane selectivity (from 21 to 19%). The H/C ratio of the total products increases from 2.28 to 2.37. The results are against the hydrogen-transfer mechanism. Another pathway is the dehydroaromatization route, which mainly occurs at Lewis acid sites.²⁸ In order to clarify the most likely routes, the relationships between the number of different acid sites and the hydrocarbon formation rates are systematically studied. It is clearly observed in Figure S14 that no correlations between the olefin and aromatic formation rates and the number of Brønsted acid sites are observed. On the other hand, the olefin formation rate decreases linearly with the increase in the number of Lewis acid sites, whereas the aromatic formation rate displays an opposite linear relationship, as shown in Figure 7. It indicates that the conversion of olefins into aromatics follows dehydroaromatization path rather than the hydrogen transfer path. Based on this mechanism, the catalytic performance over the tandem catalysts is in line with the change of acidic properties over treated ZSM-5. As the number of Lewis acid sites of ZSM-5 drops upon low temperature treatment, the ability of olefin dehydroaromatization decreases, therefore obtaining a low aromatic selectivity. Additionally, there exists other reactions in this catalytic system, like alkylation and hydrocracking reactions which mainly occur over Brønsted acid sites. These reactions would also affect the aromatic selectivity. Thus, with the decline in the number of Brønsted acid sites under treatment in high temperature, the impact of alkylation on the product distribution might become weak, further causing a higher alkane selectivity.

3.4. Proposed Reaction Paths. Given all of the above discussions, the possible reaction paths for the conversion of syngas over Na-Fe-ZrO₂/ZSM-5 are deduced, as illustrated in Figure 8. CO would first hydrogenate over iron carbide of Fe-

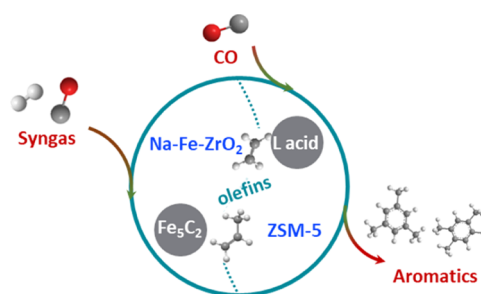


Figure 8. Proposed reaction routes in the Fe-based catalyst–zeolite process.

based catalyst to form low-carbon olefins and other hydrocarbons. Subsequently, the olefins transfer toward ZSM-5. The C₂₋₄ alkenes undergo oligomerization into C₆⁺ alkenes, which could also be produced by the alkylation of low-carbon alkanes and alkenes on Brønsted acid sites.⁵³ C₆⁺ alkenes subsequently mainly follow a dehydroaromatization route into aromatics over Lewis acid sites.^{54,55} In the meantime, CO would also incorporate into aromatics in this process by forming carbonyl groups and coupling with olefins. Accordingly, the proper addition of zeolite not only enhances the carburization degree of iron species but also offers reaction sites for olefin aromatization and CO coupling reactions. Both play a favorable role in the CO conversion and aromatic formation.

4. CONCLUSIONS

In conclusion, the combination of Na-Fe-ZrO₂ and ZSM-5 offers an alternative way for the direct conversion of syngas into aromatics. The reaction pathways of syngas to various products have been elucidated by investigating the relationship between product distribution and the CO conversion. The light olefins are determined as intermediates during the tandem process, leading to the formation of aromatics over ZSM-5. Additionally, the proper presence of ZSM-5 would enhance the iron carburization and also facilitate the incorporation of CO with olefins into aromatics. Both are favorable for achieving high catalytic activity and aromatic production. Through exploring the relationship between the number of different types of acidic sites and product distribution, it is found that the olefins mainly follow the

dehydroaromatization route to form aromatics over ZSM-5. The present work explores the interaction between Fe-based catalyst and ZSM-5, also proposes the reaction paths during this consecutive process, promoting a better understanding of the tandem catalyst system.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.9b05449>.

Catalyst characterization, including SEM images, XRD, EDS mapping, Mössbauer spectroscopy, H₂-TPR, N₂ adsorption isotherm, CO-FTIR, acid amounts, and some catalytic results (PDF)

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Notes

The authors declare no competing financial interest.

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